

>> Doctor Performance Analysis-

Objective:

The objective of this analysis is to evaluate doctor performance by examining appointment workload, consultation completion, and specialization-wise distribution. This analysis helps assess resource utilization, identify top-performing doctors, and support informed operational planning within the hospital.

Analysis Performed:

The following analyses were conducted using Python and Pandas:

- Total number of doctors
- Doctor-wise appointment count
- Completed appointments by each doctor
- Top-performing doctors based on appointment volume

Key Performance Indicators (KPIs):

KPI	Value
Total Doctors	10
Total Appointments	50
Top Doctors	Dr. Arjun Mehta, Dr. Neha Kapoor, Dr. Rohan Iyer, Dr. Priya Nair, Dr. Vikram Singh (5 appointments each)

Results:

- The hospital employs **10 doctors** across multiple medical specializations.
- Every doctor handled **5 scheduled appointments**, indicating an evenly distributed workload.
- Five doctors completed **4 consultations** each, while the remaining five completed **3 consultations** each.
- Based on appointment volume, the top-performing doctors are:
 - Dr. Arjun Mehta (Cardiologist)
 - Dr. Neha Kapoor (Cardiologist)
 - Dr. Rohan Iyer (Neurologist)
 - Dr. Priya Nair (Neurologist)
 - Dr. Vikram Singh (Orthopedic)

Key Findings:

- The hospital has a workforce of **10 doctors** representing multiple specialties.
- Appointment allocation is evenly distributed, with each doctor managing **5 appointments**.
- Consultation completion varies slightly, with half of the doctors completing four consultations and the remaining half completing three.
- Cardiology, Neurology, Orthopedics, Pediatrics, and Emergency Medicine are represented within the dataset, supporting a diverse range of healthcare services.

Business Insights:

Doctor workload is distributed evenly across the medical staff, indicating effective appointment scheduling and balanced utilization of hospital resources. This balanced allocation minimizes the risk of overburdening individual doctors while ensuring equitable distribution of patient consultations.

Although appointment volume is identical across all doctors, differences in completed consultations highlight minor variations in consultation outcomes, likely influenced by

appointment status such as no-shows or cancellations. Monitoring these metrics can help hospital administrators evaluate scheduling efficiency and identify opportunities to improve consultation completion rates.

The Top Doctors KPI identifies the leading doctors based on appointment volume and serves as a valuable dashboard metric for monitoring physician activity and operational performance.

Conclusion:

The Doctor Performance Analysis demonstrates effective workload distribution across the hospital's medical staff. The analysis confirms that appointments are allocated uniformly while consultation completion remains consistently high across all doctors. These findings provide meaningful insights into workforce utilization and establish a strong analytical foundation for monitoring doctor performance through the Power BI dashboard.